"Generate a cartoon picture to use for the introduction to my HCI class. Include scene where students in Bangladesh are learning together how to carry out a survey." [ChatGPT]

CSE472: HCI – Human Computer Interaction

Lecture – 3: Survey Design

Spring 2026

BRAC UNIVERSITY

Inspiring Excellence

Today: Survey Design

- What are surveys and types of surveys
- Cases, populations, samples
- Sampling methods and sample sizes
- Response rates
- Question design

Why Survey?

1. Reach MANY Users Quickly
2. Get Quantifiable Data
3. Understand User Preferences & Demographics
4. Validate or Test Hypotheses
5. Complement Other HCI Methods
6. Measure User Satisfaction
7. Cost-Effective
8. Track Changes Over Time

Survey

- Surveys are a method of *gathering information* from *a sample of people*, traditionally with the intention of *generalizing the results to a larger population*.

Three types of Surveys

1. Self-administered questionnaires Including:
    1. Mail(ed) surveys (or e-mail surveys)
    2. Web-based surveys
    3. Group surveys (e.g. In a classroom)
2. Interview surveys ('face-to-face')
3. Telephone surveys

Online tools for running Surveys

Time as a dimension in survey research

- Longitudinal studies: Permit observations of the same population or phenomena over an extended period of time.
- Enable analysis of change. May also facilitate more credible assertions relating to causality.

1. Trend studies

   – examine change in a population over time (e.g. Census).

2. Cohort studies

   – examine specific subpopulations or cohorts (often, although not necessarily, the same individuals) e.g. a study might interview people aged 30 in 1970, 40 in 1980, 50 in 1990, etc.

3. Panel study

   – examine the same set of people each time (e.g. the same sample of voters every month during an election campaign).

Thinking about what you're researching: Case, Population, Sample

- Case: each empirical instance of what you're researching
- So if you're researching *scientists who were nominated but never won a Nobel Prize*, Rosalind Franklin would be a case, as would Satyendra Nath Bose, Lise Meitner etc.
- If you were interested in *Fast Food companies* Pizza Hut would be a case, Burger King would be a case, as would BFC, KFC, etc.

Thinking about what you're researching: Case, Population, Sample

- Population – all the theoretically-relevant cases (e.g. 'All Team Bangladesh supporters'). This is also often referred to as the target population.
- This may differ from the study population, which is all of the theoretically-relevant cases that are actually available to be studied (e.g. 'people who have purchased tickets in 2024').

Choosing a Study Population

- Are they relevant to the problem?
- Are they the group affected by the problem?
- Or are they the group that will have to carry out the recommendations?
- Or are they the people who have dealt successfully with this problem elsewhere?
- Can you get a fair sampling? How?
- Will you get a high response rate?

Choosing a Study Population

- Sometimes you can study all possible cases (i.e., the entire population)

  e.g., All scientists who were nominated but never won a Nobel Prize

- Often you can't study the whole population because it is too big, or doing so would be too costly, too time consuming, or impossible.

  e.g., All students who ever attended BRACU

- In these cases, you need to select a sample of the population to study.

So how do you do sampling?

Rely on available participants

- Literally choosing people because they are available

  (e.g. approaching the first five people you see after class)

- Only justified if less problematic sampling methods are not possible.

- Researchers must use considerable caution in generalizing from their data when using this method

So how do you do sampling?

Snowball sampling

- Researcher collects data on members of the target population s/he can access, and uses them to help locate other members of the population.
- May be appropriate when members of a population ar e difficult to locate (and/or access).
- By definition, respondents will be connected to other respondents, thus are more likely to share similarities with each other than with other members of the population.

So how do you do sampling?

Random sampling (Representative)

- Feasible only with the simplest sort of sampling frame (a comprehensive one).
- The researcher enumerates the sampling frame (i.e. all possible participants), and randomly selects people
- It's one of the most reliable ways to get a **fair and unbiased sample** because it avoids favoritism or patterns in selection.

So how do you do sampling?

Stratified sampling

- Rather than selecting a sample from the overall population, the researcher selects cases from homogeneous subsets of the population (e.g. random sampling from a set of CM students, and from a set of MBA students).
- This ensures that key sub-populations are represented adequately within the sample.
- A greater degree of representativeness in the results thus tends to be achieved, since the (typical) quantity of sampling error is reduced.

So how do you do sampling?

Convenience sampling

- It is a non-probability sampling method where researchers select participants based on their easy accessibility and proximity, rather than a systematic or random selection.
- You choose whoever is most **convenient** for you to survey – for example, people nearby, classmates, or online users who respond quickly.
- It's fast, simple, and inexpensive – but it can introduce **bias** because the sample might not truly represent the whole population.

Sample Size

The sample size that is needed depends upon:

- The heterogeneity of the population: the more heterogeneous, the bigger the sample needed
- The number of relevant sub-groups: the more sub-groups, the bigger the sample needed
- The frequency of a phenomenon that you are trying to detect: the lower the frequency that it occurs, the bigger the sample needed
- How accurately you want your sample statistics to reflect the population: the greater accuracy that is required, the bigger the sample needed.
- How confident you want to be about your results!

Other considerations when you are thinking about sample size

- The response rate – if you think that a lot of people will not respond, you need to sample a larger number of people.
- Form of analysis – some forms of statistical analysis require a larger number of cases than others. If you plan on using one of these you will need to ensure that you've got enough cases.

- Generally (given a choice): Bigger is better!

(hence the sample size often reflects costs/resources.)

Questionnaire Design

- *Title & Purpose*
- Give your questionnaire a topic-related title
- Include a brief explanation of the survey's purpose (short paragraph)
- Include instructions for completing the survey
- Include instructions for returning the survey
- Assure respondents the survey is anonymous

Question Design

- Survey questions should be easy to understand and hard to misinterpret.
    - Use common language & clearly defined terms
    - Use mostly closed questions
    - Include demographic questions to help identify the sample
    - Use precise, unambiguous language
    - Make sure questions are valid & reliable

What to Avoid?

- Avoid double-barreled questions
    - Do you like running and going to the gym?
- Avoid leading, biased questions
    - Will this product help you to be efficient and better at your job?
- Avoid loaded questions
    - Have you stopped stealing food from the TAs?
- Avoid repetitive questions
    - Why? Why? Why? Why?
- Avoid personal questions
    - What type of relationship did you have with your parents?

Gathering Demographic Questions

- Do you expect age or gender to play a role in people's opinions?
- Do you expect education level or income level to play a role?
- Do you expect size of household or length of employment to play a role?
- Do you want to compare responses from different groups?

Sequencing your Questions

- Group questions that are similar
- Put them in a logical order
- Place demographic questions at the beginning
- Put any sensitive or difficult questions at the end
- Put any open-ended questions at the end

Types of Questions: Yes/No

Are you currently employed?

___ yes

___ no

Types of Questions: True/False

Your store is located downtown.

___ True

___ False

Types of Questions: MCQ

What is your favorite browser?

- Chrome
- Mozilla
- Firefox
- Internet Explorer
- Safari
- Other ________________

Types of Questions: Agree/Disagree Scale

Drug use is a problem among our employees.

- Strongly agree
- Agree
- No opinion
- Disagree
- Strongly disagree

Types of Questions: Ranking

How much of your shopping and recreation do you do in Chittagong?

- *Little or none*
- *Les than half*
- *About half*
- *More than half*
- *Most or all*

Types of Questions: Rating

Rate the following factors according to their importance on your choice of where to live. (5 = very important and 1 = unimportant)

- Commuting time
- The apartment size
- Rent price
- Proximity to friends and family
- Restaurants and night life

Types of Questions: Checklist

From what sources did you finance your business? Check all that apply.

- Bank
- Saving and loan institution
- Personal savings
- VC funding
- Other ___________________________

Formulating Response Categories

- Limit types of questions and response sets to no mor than three
- Give clear-cut answer choices
- Make response sets easy to navigate
- Make sure response categories don’t overlap
- Make sure responses cover all possibilities

Overlapping Response Categories

Example:

How many credit hours have you completed?

___1 - 25	___1 - 24
___25- 55	___25- 54
___55- 85	___55- 84
___85 or more	___85 or more

Checklist for your survey

- Have you asked everything you want answers to ?
- Will the answers help you make recommendations?
- Are your questions unbiased, clear, and to the point?
- Are they logically arranged?
- Is the questionnaire neat & uncluttered?
- Is it carefully edited?
- Can respondents complete it quickly & without difficulty?

Test run your Survey!

- *Once questionnaires with a mistake go out, it's too late to correct them! This will negatively affect your data AND has a negative effect on your credibility.*

Summary

- Strengths of survey research:
  - Useful for describing the characteristics of a large population.
  - Makes large samples feasible.
  - Flexible-many questions can be asked on a given topic.
  - Has a high degree of reliability and replicability.
  - Is a relatively transparent process.
  - Useful for measuring change over time.

Summary

- Weaknesses of survey research:
- Seldom deals with the context of social life.
- Inflexible – cannot be altered once it has begun.
- Subject to artificiality – the findings can be a product of the respondents' consciousness that they are being studied.
- Can be poor at answering questions where the units of analysis are not individual people.
- Can be weak at gathering at certain kinds of information, e.g.,
  - highly complex or 'expert' knowledge
  - people's past attitudes or behavior
  - subconscious (especially macro-social) influences
  - shameful or stigmatized behavior or attitudes

Cheat Sheet

Method	Advantages	Disadvantages	Tips to Remember
Self-completion	•Cheap •Cover wide area •Anonymity protected •Interviewer bias doesn’t interfere •People can take their time	•Low response rate (and possible bias from this) •Questions need to be simple •No control over interpretation •No control over who fills it in •Slow	•*Simplify questions* •*Include covering letter* •*Include stamped addressed response envelope* •*Send a reminder*
Telephone survey	•Can do it all from one place •Can clarify answers •People may be relatively happy to talk on the phone •Relatively cheap •Quick	•People may not have home phones/be ex-directory •You may get wrong person or call at wrong time •May be a bias from whose name is listed/who’s at home •Easy for people to break off •No context to interview	•*Because you rely totally on verbal communication – questions must be short and words easy to pronounce* •*Minimize number of response categories (so people can remember them)*
Face-to-face interview	•High response rate •High control of the interview situation •Ability to clarify responses	•Slow •Expensive •Interviewer presence may influence way questions are answered •If there is more than one interviewer, they may have different effects	•*Interviewer should be non-threatening* •*Interviewer can clarify questions, but should be wary of affecting the content* •*Aim to ask questions in a clear, standardized way* •*If the list of possible responses is long, show them to the respondent for them to read while the question is read out*

Reading

https://drive.google.com/file/d/1lDSg1Fd2eludDLm0RG-9_kToPIGhutGj/view?usp=sharing

Any questions?

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